

Newton Sythong

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Professional Summary

Software Engineering graduate (University of Otago, 2025) building full-stack, database-backed applications with secure authentication and multi-user architecture. Experienced integrating LLMs (OpenAI, OpenRouter) into React/Next.js and Java systems, backed by PostgreSQL and relational database design. Focused on reliable, tested software across the stack.

Technical Skills

- **Languages:** TypeScript, JavaScript, Java
- **Frontend:** React, Next.js, Vue.js, Swing (Java GUI)
- **Backend:** Node.js, REST APIs, Prisma, JDBI, JDBC/HikariCP
- **Databases:** PostgreSQL, Supabase, H2
- **AI:** OpenAI APIs, OpenRouter, structured LLM outputs, contextual prompting
- **Testing:** JUnit, AssertJ, Mockito
- **Tools:** Git
- **Engineering:** Authentication & Authorisation, Relational Database Design, Automated Testing, Design Patterns, Caching, Concurrency

Education

University of Otago	2022 – 2025
Dunedin	Bachelor of Science (Software Engineering)
	Minors: Information Science, Psychology

Experience

Workshop Facilitator

Remojo Tech (acquired by Digital Future Aotearoa), 2023 – 2026
Founded and trained student-led Recycle a Device (RAD) clubs at multiple schools, establishing independent, ongoing refurbishment and device-redistribution programs. Delivered nationwide hands-on laptop refurbishment workshops for schools and communities, teaching diagnostics, repair, and safe handling. Led “Fix 1 Keep 1” sessions, enabling participants to keep refurbished laptops and expand digital access for underprivileged individuals.

Laboratory Technician (Trainee CMT Technician)

Construction Sciences NZ (KIWA Group), Wellington, New Zealand, 2026

Head Instructor

Conducted laboratory and field testing of construction materials in line with New Zealand standards and QA procedures. Prepared and analysed samples, maintained technical records, and supported site investigations and equipment compliance. *Dunedin Taekwondo, 2020 – 2022*

Managed and adapted training programs for 20–30 participants across age groups and skill levels.

Project Experience

Note Pilot – AI-Powered Study Assistant

2025

Full-stack AI tool for lecture summarisation, flashcards, and practice problems.

- Designed secure multi-document PostgreSQL schema with Prisma for session state and strict user isolation
- Engineered contextual LLM prompting (OpenRouter) to deliver high-quality, reliable outputs
- Implemented session caching with race-condition safeguards to improve response reliability
- Developed responsive React/Next.js frontend with real-time feedback and clean UI/UX

Fresh Flat – Household Management Web App

2025

Full-stack platform for flatmates to manage shared pantries and generate AI-powered recipes.

- Implemented secure email/password authentication with Supabase, SSR middleware, and JWT-protected routes
- Developed invite-token flat creation/joining with admin roles and junction-table multi-user isolation
- Engineered role-based, access-level filtering across flats, including pantry data with expiry tracking
- Integrated GPT-4o-mini for structured recipe generation with enforced JSON schema parsing

Product Catalogue – Inventory & E-Commerce Platform

2024

Dual-interface inventory and e-commerce platform: Swing desktop admin GUI and Vue.js storefront, backed by a REST API and H2 database.

- Architected shared DAO layer (JDBI + JDBC/HikariCP) for consistent CRUD across GUI and backend
- Developed a feature-complete Swing admin GUI (search, image support, OVal validation) and a Vue.js storefront with persisted cart, authentication, and checkout
- Implemented secure REST endpoints with BasicAuth and comprehensive test coverage (JUnit, AssertJ, Mockito)

Awards and Achievements

- University of Otago Performance Entrance Scholarship (2018)
- Represented New Zealand at the World Taekwondo Championships (2017)